25900400 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFTQ202 - UNIT OPERATIONS IN FOOD INDUSTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Define a scraped surface heat exchanger. [U] / [CO1]
2. List two heat-sensitive products suitable for freeze drying. [K] / [CO2]
3. What are the limitations of water distillation? [U] / [CO1, CO3]
4. Why is radiation dominant in high-temperature food processes like broiling? [U] / [CO1]
5. What is regeneration efficiency in plate heat exchangers? [U] / [CO1]
6. State two advantages of tubular heat exchanger. [U] / [CO1]
7. Differenciate between freezing and chilling. [U] / [CO1]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. Recommend nanofiltration for specific food industry applications. [E] / [CO4]

9. Discuss how direct contact steam injection improves heating efficiency. [A] / [CO3]

10. Rank the applications of microfiltration in food processing. [E] / [CO4]

11. Derive the heat conduction equation in cylindrical coordinates and mention its food applications. [U] / [CO1]

12. Suggest a suitable drying method for leafy vegetables with reasons. [An] / [CO3]

13. Compare the rectifying section and stripping section. [U] / [CO1, CO3]

14. What are the applications of plate freezers in fish industry? [U] / [CO2]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

15. Compare forward feed and backward feed arrangements in multiple-effect evaporators. [An] / [CO3]

16. List five food products dried by spray drying and explain their importance. [K] / [CO2]

17. Discuss convective mass and heat transfer in drying of fruits and vegetables. [U] / [CO1]

18. Produce two examples of industries using RO [C] / [CO5]

[10x2 = 20]